Dr. Aadil Raza

Assistant Professor
Department of Physics
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Objective	To achieve excellence in teaching, research, development, and entrepreneurship in the field of Electronics, Nanoelectronics, Optical communication, and Fiber optic sensors.
Academic Qualifications	**Ph.D. in Electrical Engineering** *(2013-2019) from The Department of Electrical and Computer Engineering, COMSATS University Islamabad & Research work from The Hong Kong Polytechnic University (PolyU), Hong Kong* Ph.D. Thesis Title**:** Radio-over-Fiber Transmission Schemes for Next-Generation Wireless Access Networks. …….. **MS in Nanoelectronics and Mechanics** *(2008-2009) from The University of Sheffield & The University of LEEDS, UK* MS Thesis Title**:** Gain & Spontaneous Emission Measurements of Self-assembled Quantum Dot Lasers. …… **MSC in Electronics** *(2003-2005) from Quaid-i-Azam University, Islamabad, Pakistan.*
Awards (International collaboration)	• Development of C++ codes for UWB microwave transmitter and receiving antenna components, as well as a channel component for Optisystem Software, Optiwave System Inc., Canada, to enable the propagation of transmitted electromagnetic waves to the receiver component (2023). • Innovation and Collaborative Research Project from Optiwave Systems Inc (Canada) titled: "The development of a deep learning-based eye diagram analyzer for OptiSystem Software" (2024). • Innovation and Collaborative Research Project with Carlton University titled "Integrated LiDAR system for Autonomous Vehicles" (2025)
Awards (Indigenous)	• Outstanding Performance Award 2025 (FY 2024-25) from the Rector of COMSATS University Islamabad. • Outstanding Performance Award 2024 (FY 2023-24) from the Rector of COMSATS University Islamabad. • Awarded FYP funding (NGIRI-2022-14098) under the IGNITE program for the project titled "Development of a system for monitoring the real-time performance of vehicles with fault tolerance". • Awarded FYP funding (NGIRI-2022-14410) under the IGNITE program for the project titled "Experimental Study of Modulation Dynamics of Laser and Photodetector to Realize Visible Light Communication".

Research Interests	I am currently working in the following areas: • Optical Fiber Sensors • Free Space Optics • Radio over Fiber Communications • High-speed Datacenter Interconnects • MIMO and Millimeter- Wave Communication • Optical Communication Networks • Machine Learning for Optical Communication
Titles of Project Supervised	• Remote Controlled Robot Arm on Mecanum Wheels with Camera (2019-2020). • Home Automation with Modern Facilities (2019-2020). • Synthesis of Nickel Oxide Nanofibers for Gas Sensing Applications (2020-2021). • Experimental Study of Modulation Dynamics of Laser and Photodetector to Realize Visible Light Communication (2020-2021). • Development of a system for monitoring the real-time performance of vehicles with fault tolerance (2021-2022). • Synthesis of Cobalt Doped Manganese Dioxide Nanostructures for Supercapacitor (2021-2022). • A Study of Electrospun Nanofibers for Lithium Ion Battery (2021-2022). • Design and Development of a Remote Monitoring Station Using Drones and Free Space Optics for Disaster Zone Communication (2024-2025)
Experience	**Assistant Professor** (2007 to date) at Department of Physics, COMSATS University Islamabad, Pakistan. I am involved in teaching, labs, research, and any additional duties (member of various departmental committees) assigned by HOD. **Incharge** of Bachelor of Electronics (BEL) program (2023/2024). **Courses I can teach:** Electric circuit analysis, Electronics, Electromagnetic Theory, Signals and Systems, Analog Communication Systems, Digital Communication Systems, Digital Logic Design, Optical fiber communication, Fiber Optic Sensors, and any other related course. **International:** Developer at Optiwave design software (https://optiwave.com/), Canada, and technical content writer for learning management system (self-directed learning) for The TEA Team, Canada.
Research Publications	**Articles Published in Impact Factor (IF) journals.** **1.** Kangping Zhong, Changjian Guo, **Aadil Raza**, Xian Zhou, Jiahao Huo, Tao Gui, Alan Pak Tao Lau and Chao Lu., "Techniques for Performance Evaluation of Optical Transmission System and Module," **2017** Asia Communications and Photonics Conference (ACP), 2017, pp. 1-1. **2.** **Raza, A**., Ghafoor, S. & Butt, M.F.U. "MIMO-enabled integrated MGDM–WDM distributed antenna system architecture based on plastic optical fibers for millimeter-wave communication", Photon Network Communication 35, 265–273 (**2018**). https://doi.org/10.1007/s11107-017-0741-9 **3.** **Aadil Raza**, Kangping Zhong, Salman Ghafoor, Saeed Iqbal, Muhammad Adeel, Shahid Habib, Muhammad Fasih Uddin Butt, and Chao Lu, "SER estimation method for 56 GBaud PAM-4 transmission system," Chin. Opt. Lett. 16, 040604- (**2018**) **4.** Muhammad Adeel, Chao Shang, Diya Hu, Huan Wu, Kun Zhu, **Aadil Raza**,

and Chao Lu, "Impact-Based Feature Extraction Utilizing Differential Signals of Phase-Sensitive OTDR," J. Lightwave Technol. 38, 2539-2546 (**2020**)

5. Iqbal. S, **Raza. A**, Fasih Uddin Butt. M, Ghafoor. S, El-Hajjar. M. "A full-duplex radio over fiber architecture employing 12 Gbps 16 × 16 optical multiple input multiple outputs for next-generation communication networks", Trans Emerging Tel Tech. **2020**; 31: e3910.
6. Iqbal. S., Iqbal. M., **Raza. A***., Mirza. J., Ghafoor. S., El-Hajjar. M. and Butt. M., "Linearly polarized modes enabled PAM-4 data transmission over few-mode fiber for data center interconnect", Electron. Lett., 56: 1125-1127, (**2020**).
7. Jawad Mirza, **Aadil Raza**, Ahmad Atieh, Saeed Iqbal, and Salman Ghafoor "Self-restorable intra data center interconnect based on multimode fiber and free-space optics," Optical Engineering 60(3), 036113 (31 March **2021**).
8. Iqbal. S., **Raza. A**., Butt. M.F.U. et al. "Millimeter-wave enabled PAM-4 data transmission over hybrid FSO-MMPOF link for access networks", Opt Rev 28, 278–288 (**2021**).
9. Jawad Mirza, Abdulah Jeza Aljohani, **Aadil Raza**, Saeed Iqbal, Salman Ghafoor, A multi-hop free space optical link based on a regenerative relay, Alexandria Engineering Journal, Vol 61, Issue 2, 1459-1467, **2022**, ISSN 1110-0168,
10. Salman Ghafoor, Sher Afraz, **Aadil Raza** and Muhammad Fasih Uddin Butt, "Electroabsorption Modulator-Based Relay for the Transmission of DPSK-Modulated Signals over the Free Space Optical Link". Arab J Sci Eng (2022).
11. Muhammad Adeel, Javier Tejedor, Saeed Iqbal, Muhammad Muaz, **Aadil Raza*** and Javier Macias-Guarasa, "Differentiating trace-to-trace noise effects using novel signal characteristics in phase-sensitive OTDR systems" Opt Quant Electron 55, 49 (**2023**).
12. Z. Ali, K. N. Qureshi, A. S. Al-Shamayleh, A. Akhunzada, **Aadil Raza** and M. F. U. Butt, "Delay Optimization in LoRaWAN by Employing Adaptive Scheduling Algorithm with Unsupervised Learning," in IEEE Access, vol. 11, pp. 2545-2556, **2023**.
13. Areesha Ijaz, Basit Raza, Iqra Kiran, Abdul Waheed, **Aadil Raza**, Habib Shah, and Sulaiman Aftan, "Modality Specific CBAM-VGGNet Model for the Classification of Breast Histopathology Images via Transfer Learning," in *IEEE Access*, vol. 11, pp. 15750-15762, 2023.
14. **Aadil Raza*,** Saeed Iqbal, Muhammad Iqbal, Jawad Mirza, Salman Ghafoor, and Ahmad Atieh5, "400 Gbps/λ PAM-4 Data Transmission over FSO Link by Employing Space Division Multiplexing for Data Center Interconnects Using LG Modes Enabled VCSELs," Optical and Quantum Electronics, **2023**.
15. Mohammad Kaleem, Sajid Nazir, Shaista Jabeen, Hamid Turab Mirza, Shahrukh Agha, **Aadil Raza**, "Sputtering Al2O3 enhanced bandgap engineering for integrated photonic devices", Optics & Laser Technology, Volume 162, **2023**,109287.
16. Muhammad Adeel, **Aadil Raza***, Muhammad Moaz, Distributed Optical Fiber Sensing in Railway Engineering, In book: New Research on Railway Engineering and Transport, Publisher: IntechOpen, **2023** (**Book Chapter**)
17. Iqbal, S., **Raza. A**., Kaleem, M., et al. A full duplex LG modes enabled

	millimeter-wave based FSO communication system for the disaster zone. Wireless Network (2023). **18.** Mirza, J., **Raza. A**., Atieh, A. et al. Performance Enhancement of Er–Yb: Co-doped Waveguide Amplifier Employing Backward Pumping in the Presence of Energy Transfer Up Conversion. Arab J Sci Eng (2023). **19.** Aslam, M. Z., Raza, B., Faheem, M., & **Raza. A**. AML-Net: Attention-based multi-scale lightweight model for brain tumor segmentation in internet of medical things. CAAI Transactions on Intelligence Technology, 1–17 (2024). **20.** A. S. Al-Shamayleh, **A. Raza***, Z. Ali, S. Malik, S. Iqbal, B. Raza, M. Iqbal, and A. Atieh "Performance Evaluation of Optical Transmission Based on Link Estimation by Using Deep Learning Techniques," in IEEE Access, vol. 12, pp. 64126-64139, 2024. **21.** Fiaz A, Raza B, Faheem M, **Raza. A**. A deep fusion-based vision transformer for breast cancer classification. Health Technol Lett. 2024 Oct 23;11(6):471-484

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